

Small Sample Sizes:

So far, we've built CI's for the mean of

- Any population distribution
- With known or unknown variance

|| • Only when the sample size is large ($n > 30$ for known variance, $n > 40$ o/w)

↑ We will be able to remove this restriction, but only by adding another - our population will need to be normal.

⊛ Prop. (Distribution of $\bar{X}_n$ for Normal Pop. w/ known variance):

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, with known σ^2 .

Then, for any n :

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim N(0, 1), \text{ exactly.}$$

Proof: This follows from the fact that a linear combination of normal random variables will always be normal.

Since we are dividing by σ , a constant, this holds.

Note: 3 differences from the CLT:

- i) This holds for any n
- ii) The resulting distribution is exact
- iii) This applies only when X_i 's are iid Normal

So, we can now build CI's for the mean of a normally distributed population when the variance is known, for any sample size.

Corollary (CI's for pop. mean - known variance, normally distributed):

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, where μ is unknown, but σ^2 is known. Then, for any n :

- $[\bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}]$ is an exact two-sided $1-\alpha$ CI for μ
- $[\bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha}, \infty)$ is an exact lower one-sided $1-\alpha$ CI for μ
- $(-\infty, \bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha}]$ is an exact upper one-sided $1-\alpha$ CI for μ

Note: A subtle difference here is that these confidence intervals are exact, since $\textcircled{*}$ above, unlike the CLT, gives the exact distribution of $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$.

Now, we may be tempted to just replace σ , when unknown, with $s^2(X_1, \dots, X_n)$ or another point estimator for the population variance.

Q: If $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim N(0,1)$, then is $\frac{\sqrt{n}(\bar{X}_n - \mu)}{s(X_1, \dots, X_n)}$ also std. normal?

A: No. Definitely not for small values of n .

Why?

→ Instead of dividing by the constant σ , we are dividing by the random variable $s(X_1, \dots, X_n)$. This makes the latter standardized sample mean, $\textcircled{*}$, noisier.

Still, the CLT tells us that as long as n is large enough, we should be ok to approximate $\textcircled{*}$ as standard normal.

→ What about for small n ?

Thankfully, it is actually possible to derive the exact distribution of $\hat{\theta}$ for small values of n !

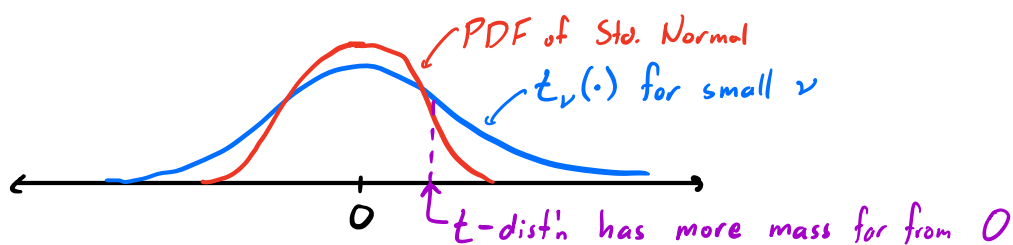
Def. (Student's t -distribution): Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Then, for any n , the statistic

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S(X_1, \dots, X_n)} \quad \text{(*)}$$

has a Student's t -distribution (t -distribution, for short) with $n-1$ degrees of freedom.

Properties of t -distribution:

- Denote by $t_\nu(\cdot)$ the pdf of a r.v. with a t -distribution with ν degrees of freedom.
- Compared to the pdf of a std. normal r.v., $t_\nu(\cdot)$ is still bell-shaped and centered at 0, but more spread out (i.e. thicker tails).



- As the degrees of freedom increase, $t_\nu(\cdot)$ becomes less spread out, and as $\nu \rightarrow \infty$, $t_\nu(\cdot)$ becomes the standard normal pdf.

→ A good sanity check, o/w this conflicts with the CLT.

- For those curious,

$$t_\nu(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \cdot \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}},$$

where $\Gamma(\cdot)$ is the gamma function.

You will not be expected to know any of this...

Since the distribution of $\bar{X}$ is not standard normal, we can't use critical values z_α of a std. normal distribution to build confidence intervals.

$$\leadsto P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{S(X_1, \dots, X_n)} \geq z_\alpha\right) \neq \alpha, \text{ (in fact, it is } \geq \alpha \text{)}$$

especially for small n .

Def. (t-critical values): If T has a t -distribution with ν degrees of freedom, then for any $\alpha \in (0, 1)$, $t_{\alpha, \nu}$ satisfies:

$$P(T \geq t_{\alpha, \nu}) = \alpha.$$

Observations: Similar to the critical values of a std. normal distribution, and since the t -distribution is symmetric, we have:

- $P(T \leq t_{\alpha, \nu}) = 1 - \alpha$
- $P(T \geq -t_{\alpha, \nu}) = 1 - \alpha$
- $P(-t_{\frac{\alpha}{2}, \nu} \leq T \leq t_{\frac{\alpha}{2}, \nu}) = 1 - \alpha$

$\leadsto$ We can use these to build CI's!

Note: Common critical values for some values of ν .

ν	10%	5%	2.5%	1%
1	3.1	6.3	12.7	31.8
5	1.5	2.0	2.6	3.4
10	1.4	1.8	2.3	2.8
20	1.3	1.7	2.1	2.5
∞	1.28	1.65	1.96	2.32

$t_{\alpha, \nu}$

z_α

Note- larger than z_α for $\alpha < 1/2$. Due to heavier tails!

More similar than for small ν

Prop. (CI's for pop. mean - unknown variance, normally distributed):

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, where μ and σ^2 are unknown.

Then, for any n :

- $\left[\bar{X}_n - \frac{s}{\sqrt{n}} t_{\frac{\alpha}{2}, n-1}, \bar{X}_n + \frac{s}{\sqrt{n}} t_{\frac{\alpha}{2}, n-1} \right]$ is an exact two-sided $1-\alpha$ CI for μ

- $\left[\bar{X}_n - \frac{s}{\sqrt{n}} t_{\alpha, n-1}, \infty \right)$ is an exact lower one-sided $1-\alpha$ CI for μ

- $(-\infty, \bar{X}_n + \frac{s}{\sqrt{n}} t_{\alpha, n-1}]$ is an exact upper one-sided $1-\alpha$ CI for μ

where $s^2(X_1, \dots, X_n) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.

Example:

Let $X_1, \dots, X_{21} \sim N(\mu, \sigma^2)$, with μ and σ^2 unknown.

Realizations $x_1, \dots, x_{21}$ yield $\bar{x} = 82$, $s^2 = 144$.

Find a (two-sided) 95% CI for μ :

First: $t_{2.5\%, 20} = 2.1$, so $\frac{s}{\sqrt{21}} \cdot 2.1 \approx 5.5$

Hence: $[82 - 5.5, 82 + 5.5] = \underline{\underline{[76.5, 87.5]}}$ is a 95% CI.

Note: If we had used critical values of the standard normal distribution, $z_{2.5\%} = 1.96$, then the CI for μ would have been narrower.

Hence, we would be less confident that this interval contains the population mean.

$\Rightarrow$ Small-sample CI's, using the t -distribution, are larger!

Hypothesis Testing:

This is our third major unit of inference.

Recall:

- Point Estimation — Aims to give one sensible value for a population parameter.
- Confidence Sets — Aims to give a range of likely to contain a population parameter.

In **Hypothesis Testing**, our goal is to answer Yes/No, as to whether a particular statement is supported by the data.

→ We call such statements **statistical hypotheses**.

Def. (Statistical Hypotheses): A claim about:

- i) the value(s) of an unknown population parameter
- ii) the form of a population distribution. || We will not see this kind of hypotheses in this class.

Examples of Hypotheses:

- i) [• "The probability of this coin landing heads-up is $1/2$ "
- ii) [• "The returns of the S&P are normally distributed"

We will test these hypotheses by seeing how strongly our observed data supports (or contradicts) them.

Motivation: Coin Flip

Let $X_1, \dots, X_{10} \stackrel{iid}{\sim} \text{Ber}(p)$, with unknown p .

- $X_i = 1$ means the i^{th} flip was a Heads

Hypothesis: $p = 1/2$.

Consider whether you (intuitively) believe our hypothesis is true if we actually observe:

- ★ • All 10 flips are "H" $\rightarrow$ Very unlikely outcome if $p = 1/2$ (has prob. 0.1%)
- 6 flips are "H" $\rightarrow$ Somewhat likely outcome if $p = 1/2$ (prob. 21%)
- 5 flips are "H" $\rightarrow$ Somewhat likely outcome if $p = 1/2$ (prob. 25%)

But now imagine we flipped this coin 100 times:

- ★ • All 100 are "H" $\rightarrow$ If $p = 1/2$, occurs w/ prob. ≈ 0
 - ★ • 60 "H"s $\rightarrow$ If $p = 1/2$, occurs w/ prob. $\approx 2 \times 10^{-4}$
 - 50 "H"s $\rightarrow$ If $p = 1/2$, occurs w/ prob. $\approx 8\%$
- || Come from the p.m.f. of a Binomial $(100, 1/2)$ r.v.

★: These outcomes really seem to contradict our hypothesis that $p = 1/2$, since we're very unlikely to see these outcomes "by chance" if p did equal $1/2$.

$\rightarrow$ Hence we might reject the hypothesis that $p = 1/2$

Takeaways:

1) We think of rejecting a hypothesis if our observations seem incompatible (i.e. very unlikely) with the hypothesis.

2) Whether we reject a hypothesis depends on the sample size and the realizations we obtained.

$\rightarrow$ Before we obtain realizations, to reject (or not) is random.

[Think of a point estimate $\hat{\theta}(x_1, \dots, x_n)$ vs. an estimator $\hat{\theta}(X_1, \dots, X_n)$]

We'll now try to formalize these two ideas.
